Manwe Castro

manwe123@asu.edu | (602) 350-3164 | linkedin.com/in/manwe-castro | https://manwe-123.github.io/

Education

Bachelor of Science in Mechanical Engineering Tempe, AZ
Arizona State University | GPA: 3.61 | Dean's List: Fall 2024, Spring 2025, Fall 2025 *May 2028*
Relevant Coursework: Mechanics of Materials, Structure and Properties of Materials, Thermodynamics, Fluids, Circuits

Skills

Design: SolidWorks, Fusion 360, LTSpice, 2D Drafting, TinkerCAD
Fabrication: FDM 3D Printing, Circuit Wiring. Laser Cutting, Orca Slicer, Automotive Repair
Programming: Python, Arduino IDE (C++), Unity Game Engine, C#, Git, GitHub, Swift, MATLAB, Sensor Integration
Personal: Sewing, Violin, Guitar, Piano, Jarana, Requinto, Boxing, Rockclimbing, Photography, Salsa

Professional Experience

Institute for Digital Inclusion Acceleration Chandler, AZ
Digital Navigator *January 2025 – Present*

- Instruct over 300 weekly visitors in hardware and software fundamentals, facilitating thousands of hands-on technology interactions across diverse age groups.
- Develop and deliver structured curriculum for nine distinct software platforms, including **Swift**, **TinkerCAD**, and **SpheroEDU**, standardizing the instructional process for the center.
- Operate and maintain Flashforge 3D printers using **Orca Slicer**, troubleshooting hardware failures and optimizing print parameters to ensure consistent output quality.

Project Experience

Mental Health Robotics System for Isolated Environments, Arizona State University Tempe, AZ
Design and Prototyping Lead *August 2024 – December 2024*

- Programmed an **Arduino Uno** to process **ultrasonic** and **photoresistive sensor** inputs, controlling drive motors and a servo actuator to enable autonomous user tracking and automated medical dispensing, achieving 95%+ proximity detection
- Modeled a 1.8 cubic foot robotic enclosure in **SolidWorks** and fabricated the structure, integrating sensors, motors, display hardware, and battery components into a prototype build, finishing 40% under budget
- Executed formal acceptance testing across functional requirements, diagnosing and resolving an electrical power delivery fault to pass sensor, motor, and dispensing tests, validating semi-autonomous navigation and automated medical access.

VR Rapid Prototyping Application, EPICS at Arizona State University Tempe, AZ
Prototyping Lead | Design Team *March 2025 – December 2025*

- Engineered a virtual reality prototyping environment for Meta Quest headsets using **Unity and C#**, integrating over 100 interactive 3D assets with simulated physics and collision mechanics to support rapid design across 30+ headsets
- Directed a cross-functional design team by establishing standardized Unity workflows and resolving engine-level rendering blockers, optimizing frame rates to mitigate VR motion sickness
- Collaborated with faculty to integrate a 2-week curriculum into the experience and developed in-game analytics scripts to track completion time, errors, and hints, creating a data-driven feedback loop for instructors.
- Authored Gantt charts, risk matrices, and standardized technical reports for a 12-person, multi-semester project, delivering a transition package that enabled future teams to continue development, testing, and classroom deployment.

Shock Absorber Assembly, Arizona State University *August 2025 – December 2025*

- Designed a fully constrained automotive shock absorber assembly in **SolidWorks** by modeling custom components, integrating hardware, and amending dimensions to eliminate physical interferences for dynamic suspension applications.
- Generated fully dimensioned **2D engineering drawings** following **ASME Y14.5 standards**, detailing complex features such as shaft thread specifications, keyways, and spring parameters to provide precise fabrication instructions.
- Authored a comprehensive **Bill of Materials** and exploded assembly drawing to document custom and standard hardware, establishing a standardized technical reference to streamline part procurement.

Leadership and Campus Involvement

Society of Hispanic Professional Engineers, ASU Chapter Tempe, AZ
Outreach Director *March 2026 – Present*

- Direct chapter outreach by partnering with 15 local schools and 6 community organizations, prioritizing schools with large Hispanic student populations.
- Plan and coordinate outreach and professional development events of up to 60 attendees including resume workshops and Noche de Ciencias
- Manage a 4-member student events committee, 15+ volunteers, and a $1200 annual budget while designing hands-on technical workshops throughout the semester that build practical engineering skills for undergraduate members.